Miguel Aréjula Aísa

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Education

Master in Software Engineering, University of Southern Denmark — June 2027

- **Coursework:** Advanced Software Engineering Methodologies, Advanced Software Architecture and Analysis Techniques, Big Data and Data Science Technology

BSc in Computer Science, University of Zaragoza — June 2021

- **Coursework:** Software Engineer, Distributed Systems, Software Architecture, Prerequisite Engineering, Verification and Validation, Agile Methodologies and Quality, and Artificial Intelligence.
- Achieved high honors in Information Systems II and Verification and Validation, demonstrating exceptional proficiency and understanding in these subjects.

Cambridge English Advance, Cambridge — July 2024

Experience

Software Engineer, University of Zaragoza — 2024-06-01 - 2025-08-31

- Led development of custom web application for Traumatology Department
- Developed full-stack system using React, PostgreSQL, Express, and FastAPI
- Integrated clinical data into relational model for clinical pathway validation

Teaching Assistant, University of Southern Denmark — 2026-02-01 - 2026-06-01

- Support students in understanding core software architecture concepts by assisting with lab sessions, assignments, and problem-solving, while clearly explaining technical concepts and collaborating with professors.

Projects

Are-Dev — Link

- Personal technical blog and YouTube channel focused on software development, front-end technologies, and modern frameworks
- Published blog posts and videos sharing tutorials, project walk-throughs, and insights to engage the developer community **Tools Used:** Astro, Vercel, Markdown.

I4 Pizza Production System — Link

- Industry 4.0 pizza production system integrating warehouse, production line, and web platform via distributed architecture
- Designed, validated, and documented system architecture using ADD with requirements, use cases, microservice boundaries, and formal verification with UPPAAL
- Led system-level coordination, ensured architectural consistency, and supervised implementation **Tools Used:** Go, Python, Kafka, Docker, UPPAAL.

AgroNet — Link

- Full-stack collaborative web platform for farmers integrating real-time market prices, personalized weather alerts, and interactive geospatial data visualization
- Led front-end team development and deployed on Vercel
- Managed backend communication hosted on AWS ensuring reliable API integration and real-time data synchronization **Tools Used:** Astro, NodeJs, TypeScript, MongoDB, AWS, Vercel.

Energy Price Prediction — Link

- Analyzed Danish Meteorological Institute weather datasets and national energy consumption/price data to forecast electricity prices
- Applied big data frameworks and machine learning models to process large-scale datasets for predictive analytics covering renewable energy impact, peak price periods, and seasonal patterns **Tools Used:** Kafka,

HDFS, Kubernetes, Hive, Spark, Python.